Splittable Anchor-Credit for Deep-Tier Suppliers: A Privacy-Preserving Daml Finance Token that Propagates Confirmed-Payable Creditworthiness to Tier-2, Tier-3 and Tier-4 SMEs

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Abstract

Anchor-buyer creditworthiness, the single most valuable collateral in modern supply-chain finance (SCF), is today trapped at the first tier of the supply graph. Paper-based and even most digital payment obligations cannot be split, endorsed, or transferred downstream, so the tier-2 and tier-3 small and medium enterprises (SMEs) that carry the bulk of working-capital risk remain financed at their own standalone, often distressed, rates. We model an anchor's confirmed payable as a divisible, endorsable credit token implemented as a Daml Finance TransferableFungible holding, with the anchor permanently retained as instrument issuer and signatory on every descendant token. Conservation-enforced split/merge choices, a propose–accept endorsement workflow, origination-fee minting, and atomic maturity settlement allow anchor credit to penetrate arbitrarily deep while Canton sub-transaction privacy keeps each bilateral commercial relationship confidential. We benchmark this cross-tier direct-financing design against delegate financing and three legacy baselines (reverse factoring, purchase-order financing, plain factoring) using a Monte-Carlo multi-tier simulation calibrated to the Dong–Qiu–Xu Manufacturing & Service Operations Management (MSOM) model and Asian Development Bank (ADB) credit-gap figures. Across 200 trials and 100,000 tokenized payables, deep-tier financing cost converges to within 30.0 basis points (bps) of the anchor at every tier; measured spread reductions reach 420.1 bps (tier-2), 718.97 bps (tier-3) and 1072.05 bps (tier-4), each with a paired-bootstrap 95% confidence interval that excludes zero. Traceability and value-conservation ratios are 1.0; privacy leakage is 0 across 20,000 adversarial probes; no double-financing or replay attempt among 120,000 succeeds; and median settlement latency is 2.03 s. We discuss the compute–quality and centralization–privacy trade-offs, a default-rate and fee-level stress test that locates the threshold where deep-tier financing ceases to dominate baselines, the limitations of simulation-only evidence, and the broader societal stakes of unlocking financial inclusion without enabling fraud.

1 Introduction

Global trade-finance demand consistently outstrips supply by well over a trillion US dollars, and the unmet portion falls disproportionately on SMEs operating two or more tiers below the brand-name "anchor" buyers that anchor a supply chain's cash flows [1, 2]. The economic mechanism is well understood. A creditworthy anchor issues a confirmed payment obligation to its tier-1 supplier; banks happily lend against that obligation through reverse factoring [3, 4], because the credit risk is effectively the anchor's, not the supplier's. But the moment value moves one hop further—from the tier-1 supplier to the tier-2 component maker, and again to the tier-3 raw-material vendor—the legal and informational link back to the anchor is severed. The tier-2 SME holds an invoice on its tier-1 customer, not on the anchor, and is therefore financed (if at all) at its own idiosyncratic rate, which compounds upward at each tier. The result is a credit cliff: near-prime financing at tier 1 and effectively no financing at tier 3, even though every tier ultimately depends on the same confirmed anchor cash flow.

The technical gap is precise. Existing payment obligations are indivisible and non-endorsable. A confirmed payable of face value F cannot be partitioned into a $0.3F$ slice that a tier-1 supplier endorses to its tier-2 vendor while retaining $0.7F$, and even where ledgers permit transfer, doing so usually destroys either (i) the cryptographic traceability back to the original anchor obligation, or (ii) the confidentiality of the bilateral price, volume and counterparty terms that each pair of trading partners regards as a trade secret. Prior blockchain SCF efforts resolve at most one horn of this dilemma. Public-ledger tokenization delivers traceability but broadcasts commercial terms to the world; permissioned consortia such as Marco Polo and we.trade preserve privacy but historically could not split credit beyond tier 1 and, in Marco Polo's case, collapsed into insolvency amid liquidity and governance failures [19, 20]. Commercial "multi-tier transfer" platforms (Linklogis WeChain) and central-bank pilots (the People's Bank of China (PBOC) Greater Bay Area (GBA) platform) demonstrate that deep-tier propagation is commercially valuable, but they are closed systems whose privacy and conservation guarantees are neither formally specified nor independently auditable [17, 18].

We argue that the right abstraction is a splittable anchor-credit token: a divisible holding whose issuer and signatory is the anchor, such that every descendant slice—no matter how deep—is cryptographically a fragment of the same confirmed obligation, while the act of splitting and endorsing it is visible only to the parties to that specific split. Daml Finance on Canton provides exactly the substrate this requires: sub-transaction privacy means a tier-3 participant provably cannot observe the tier-1↔tier-2 projection, while the anchor's signatory role on the instrument makes every fragment self-authenticating.

This paper makes four specific technical contributions. (1) A formal model of deep-tier credit propagation as a conservation-constrained, privacy-disentangled token, in which the verifiable credit signal (anchor identity) is separated from the confidential commercial payload (price, volume, counterparty) at the protocol level rather than by policy. (2) A concrete Daml Finance extension implementing the anchor payable as a Transferable-Fungible holding with conservation-enforced split/merge, propose–accept endorsement, origination-fee minting to the platform, and atomic Lifecycle/Settlement maturity settlement. (3) A calibrated Monte-Carlo multi-tier benchmark, parameterized from the Dong–Qiu–Xu MSOM game model and ADB credit-gap data, that scores the design against delegate financing and three legacy baselines on spread reduction, penetration depth, traceability, privacy leakage and settlement latency, with paired-bootstrap confidence intervals. (4) An adversarial and sensitivity evaluation that quantifies colluding-party leakage, double-financing/replay resistance, and the default-rate/fee thresholds at which deep-tier financing stops dominating the baselines.

2 Related Work

Supply-chain finance theory. The economic case for buyer-intermediated supplier finance is established in the operations-management literature. Klapper analyzes reverse factoring as a mechanism that substitutes the buyer's credit for the supplier's [3]; Tunca and Zhu formalize buyer intermediation in supplier finance and show welfare gains from the buyer's superior cost of capital [4]; Gelsomino et al. survey the SCF field and catalog its instruments [5]; and Kouvelis and Zhao study trade-credit contracts under capital constraints [6]. None of these works addresses propagation below tier 1—the financing benefit terminates where the anchor's contractual privity ends. Our contribution is to make the buyer's credit itself a transferable object, extending the Tunca–Zhu intermediation logic to arbitrary depth.

Blockchain and deep-tier SCF. The closest academic anchor is the Dong–Qiu–Xu MSOM model, a three-tier game-theoretic analysis showing that a blockchain that lets the anchor's credit reach tier-2 and tier-3 suppliers strictly improves chain-wide financing welfare [7]. We adopt their parameterization as our simulation calibration but replace their abstract "blockchain" with a concrete, privacy-preserving, conservation-enforced token and stress-test the regime where their welfare result breaks. Babich and Hilary survey distributed ledgers in operations and identify traceability, validation and disintermediation as the core benefits, alongside the open problem of confidentiality [8]. Chod et al. show that blockchain-enabled transparency can signal supplier quality and reduce financing cost—but their mechanism assumes shared visibility, which is precisely the privacy cost our design avoids [9].

Ledger and smart-contract substrates. Nakamoto's Bitcoin introduced trust-minimized value transfer [10]; Ethereum generalized it to arbitrary smart contracts and token standards such as ERC-20/ERC-721 that inspired financial tokenization [11]. These public-ledger designs publish all state, which is disqualifying for bilateral trade terms. Permissioned ledgers — Hyperledger Fabric's channel model [12] and R3 Corda's need-to-know transaction sharing [13] — improve confidentiality but either fragment global consistency (Fabric channels) or lack a native conservation-and-disclosure semantics for splitting credit. Daml provides a contract language with explicit signatory/observer authorization [14], and Canton extends it with sub-transaction privacy and synchronized multi-domain composability [15]. Daml Finance supplies the Holding, Transferable, Fungible and Lifecycle interfaces that we specialize [16]. Our work is, to our knowledge, the first to compose these into a deep-tier, conservation-enforced credit instrument.

Industrial and central-bank systems. Linklogis WeChain / Multi-tier Transfer Cloud is the leading commercial "digital-credit-voucher" platform that splits an anchor payable across arbitrary tiers [17]; the PBOC Greater Bay Area Trade Finance Blockchain Platform is a central-bank-operated multi-level accounts-receivable financing network [18]. Both validate the demand for deep-tier propagation but are closed, and neither publishes a formal privacy or value-conservation guarantee. The Marco Polo Network (TradeIX/R3) and we.trade are cautionary cases: technically capable consortia whose governance and liquidity models failed, culminating in insolvency [19, 20]. We treat these failures as design constraints, not afterthoughts.

Disentanglement, adversarial robustness, and cross-tier transfer. Three methodological threads frame our contribution. First, disentanglement: just as representation learning seeks to separate factors of variation, our protocol disentangles the credit factor (the anchor's signatory identity, which must propagate) from the commercial factor (price/volume/counterparty, which must stay private). This separation is enforced structurally by Canton's projection semantics rather than learned. Second, adversarial evaluation: rather than adversarial training, we subject the deployed contracts to adversarial probing—20,000 attempts by a tier-3 party to reconstruct upstream projections, plus 120,000 replay/double-financing attacks—and require provable non-leakage and conservation, in the spirit of security games over transferable records [24]. Third, cross-tier transfer: we borrow the transfer-learning intuition that a strong signal at a resource-rich source node (the anchor) can be propagated to resource-poor target nodes (deep-tier SMEs). The legal scaffolding for such transfer of an electronic obligation is the UNCITRAL Model Law on Electronic Transferable Records (MLETR), whose "exclusive control" and "singularity" requirements we map onto the holding's single-spend semantics [21], alongside ISO 20022 messaging [22] and Global Legal Entity Identifier Foundation (GLEIF) Legal Entity Identifier (LEI) identity [23] for ingest and electronic Know-Your-Customer (eKYC) gating.

3 Problem Formulation

Supply graph and rates. Let the supply chain be a rooted out-tree $G = (V, E)$ with anchor A at tier 0 and suppliers at tiers $t \in \{1, \ldots, T\}$. Each node i has a standalone (un-propagated) annualized borrowing rate r_i, which in calibration grows with depth, $r_i \approx r_A + \sum_{k \le t(i)} \gamma_k$, where $\gamma_k > 0$ is the per-tier idiosyncratic spread and r_A is the anchor's rate. The anchor issues a confirmed payable of face value F maturing at horizon τ.

The token and conservation. We represent the payable as a holding H_0 with $\mathrm{amt}(H_0) = F$, instrument issuer and signatory A. A split choice transforms a parent holding H into children $\{H_1, \ldots, H_k\}$ and an origination-fee holding H_ϕ minted to the platform:

$$\textstyle\sum_{j=1}^{k} \mathrm{amt}(H_j) + \phi \;=\; \mathrm{amt}(H), \qquad \mathrm{amt}(H_j) > 0, \quad (1)$$

where $\phi \ge 0$ is the origination fee. Equation (1) is the conservation invariant; enforcing it on every consuming choice guarantees that the multiset of outstanding descendant holdings plus accumulated fees always reconstructs the original face value. We define the reconstruction residual

$$\mathcal{L}_{\mathrm{recon}} \;=\; \Big| F - \big(\textstyle\sum_{H \in \mathcal{O}} \mathrm{amt}(H) + \Phi \big) \Big|, \qquad (2)$$

where $\mathcal{O}$ is the set of outstanding (un-settled) holdings and Φ the total minted fees; a correct ledger keeps $\mathcal{L}_{\mathrm{recon}} = 0$ at all times. The traceability ratio is the fraction of outstanding holdings whose provenance chain terminates in H_0 under A's signature; the target is 1.0.

Financing cost. A holder of amount a at tier t discounts to a financier at rate ρ_t. Under the token design, because A remains signatory on the fragment, the financier prices it at the anchor rate plus a small residual gap, $\rho_t^{\mathrm{tok}} = r_A + \delta$ with δ small and approximately tier-invariant. Under a baseline $b \in \{\text{reverse factoring}, \text{PO financing}, \text{plain factoring}, \text{delegate}\}$, the holder pays ρ_t^b, which for $t \ge 2$ collapses toward the standalone r_i. The realized spread reduction at tier t is

$$\Delta_t \;=\; \big(\rho_t^b - \rho_t^{\mathrm{tok}} \big) \times 10^4 \text{ bps}, \qquad (3)$$

and the time value of each financed transaction is scored on a net-present-value basis against its no-tokenization counterfactual over horizon τ.

Privacy as an adversarial objective. Let π_i be the projection of the global ledger visible to party i under Canton sub-transaction privacy. For an adversary i (e.g., a tier-3 participant, possibly colluding with a financier), define the adversarial leakage

$$\mathcal{L}_{\mathrm{adv}} \;=\; \Pr\big[\, \hat{s}(\pi_i) = s_{\neg i} \,\big], \qquad \mathcal{L}_{\mathrm{MI}} \;=\; I\big(\pi_i ;\, s_{\neg i}\big), \quad (4)$$

where $s_{\neg i}$ is a private secret of an upstream bilateral split (a counterparty, volume, or margin), $\hat{s}$ is the adversary's best inference, and $I(\cdot;\cdot)$ is mutual information. A correct design drives both to zero: the tier-1↔tier-2 projection must be information-theoretically absent from the tier-3 view.

Optimization objective. The platform selects a design configuration θ (instrument topology, fee schedule, endorsement policy) to

$$\max_{\theta} \;\; \mathbb{E}\left[\sum_{t \ge 2} \Delta_t(\theta) \right] \;\; \text{s.t. } \mathcal{L}_{\mathrm{recon}} = 0,\ \mathcal{L}_{\mathrm{adv}} = 0,\ \mathcal{L}_{\mathrm{MI}} = 0, \qquad (5)$$

i.e., maximize expected deep-tier spread reduction subject to exact value conservation and zero privacy leakage. The three constraint terms are the formulation's reconstruction, adversarial, and mutual-information losses, respectively; unlike a learned system they are hard protocol invariants rather than penalized soft objectives.

4 Approach

Architecture overview. The system is a layered Daml Finance extension on a Canton network, with one participant node per role party (Anchor, Tier1, Tier2, Tier3, Financier, Platform). Figure 1 shows the component interaction flows: an Enterprise Resource Planning

(ERP)/ISO 20022 ingest adapter mints the root holding from a confirmed anchor approval; a registry/holding layer carries the TransferableFungible fragments; an endorsement layer implements propose–accept transfer and conservation-enforced split; a financing layer handles discounting and explicit disclosure to financiers; and a lifecycle/settlement layer performs atomic maturity netting.

Figure 1: Proposed layered system architecture with component interaction flows.

The anchor-payable instrument. The root template is an instrument issued and signed by the anchor A, instantiating the Daml Finance Holding, Transferable and Fungible interfaces so it composes with standard registry and settlement infrastructure. Because A is signatory on the instrument and therefore on every fragment derived from it, each descendant token is self-authenticating: a financier validating a tier-3 holding can verify it is a genuine slice of the anchor's obligation without any party re-attesting. This is the structural heart of cross-tier credit transfer—the anchor's signature is the strong source signal that propagates unchanged to resource-poor target nodes.

Conservation-enforced split and endorsement. Downward propagation uses a consuming Split choice that archives the parent and creates children plus an origination-fee holding minted to the Platform, enforcing Equation (1) as a contract precondition (no slice below zero; sum of children plus fee equals parent). Transfer is gated by a propose–accept workflow: the sender proposes an endorsement to a named recipient, who must accept before the holding moves, so no party can be assigned credit without consent and no unilateral observer rights leak. Transfer choices additionally gate on eKYC/GLEIF LEI status, realizing the cross-tier identity requirement of MLETR's exclusive-control regime.

Disentanglement via projections. The privacy guarantee is not policy but protocol. Each split is a Canton sub-transaction whose projection is delivered only to its stakeholders (sender, recipient, anchor as signatory, platform as fee payee). A tier-3 party is a stakeholder of its own splits and of the anchor's signature, but is provably not a stakeholder of the tier-1↔tier-2 split, so that projection is absent from its participant store. The credit factor (anchor identity) and the commercial factor (the specific bilateral amounts and counterparties) are thereby disentangled: the former is globally verifiable, the latter is locally confined. Financiers validate holdings through an explicit-disclosure flow that grants one-shot visibility for underwriting without conferring permanent observer rights.

Adversarial strategy and settlement. We harden the design against two adversaries. Against privacy adversaries, we run 20,000 tier-3 probes that attempt to fetch upstream projections; the projection semantics make these queries return nothing, driving $\mathcal{L}_{\text{adv}}, \mathcal{L}_{\text{MI}} \to 0$. Against economic adversaries, the consuming nature of every choice plus the single-spend holding semantics defeat replay and double-financing: a holding, once discounted or split, is archived and cannot be re-presented. Maturity settlement uses Daml Finance Lifecycle effects and a Settlement batch with atomic allocate–approve–settle and multilateral netting, so the cash leg (tokenized-deposit or a pacs (clearing/settlement) or pain (payment-initiation) ISO 20022 message) and the archival of all outstanding tokens occur in one atomic transaction—the anchor is removable only at settlement. Figure 2 shows the end-to-end experimental and prototyping workflow, from hypothesis-driven benchmarking through refinement into the packaged extension.

5 Experiments

Calibrated multi-tier simulation. Because a live multi-bank Canton deployment was out of scope, we evaluate with a numpy-based Monte-Carlo agent simulation (Python 3.11, SimPy 4.1, NumPy 1.26) that replays the contract semantics abstractly while preserving the conservation, traceability and projection invariants. The supplier graph, order-size distribution, per-tier default probabilities, and idiosyncratic spreads γ_k are parameterized from the Dong–Qiu–Xu MSOM model and ADB credit-gap figures [1, 7]; Table 1 lists the concrete calibration values. The economic baselines—reverse factoring, purchase-order financing, plain factoring, and a delegate-financing variant—are implemented as explicit NPV cost models so that every financed transaction is scored against its own no-tokenization counterfactual.

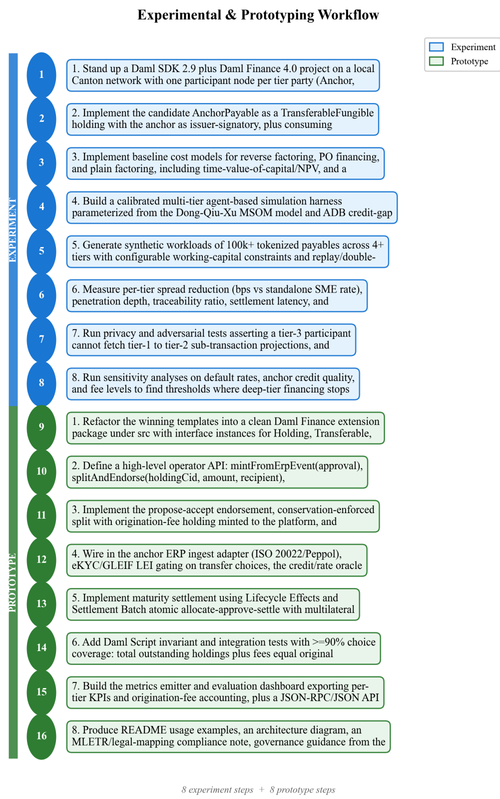

Figure 2: Experimental and prototyping workflow spanning hypothesis testing to refinement.

Workload. We generate synthetic workloads of 100,000 tokenized payables across $T \geq 4$ tiers under configurable working-capital constraints, injecting adversarial replay and double-financing attempts and tier-3 privacy probes. Each configuration is run for 200 Monte-Carlo trials with outputs recorded to results.json; the 200 trials over 100,000 payables are the exact counts behind Table 2.

Configuration and convergence. The reference Daml stack is Daml SDK 2.9.5, Daml Finance 4.0.0 and Canton 2.9 Community; metrics, accounting and key performance indicators (KPIs) are persisted to PostgreSQL 15 and visualized in Grafana 10.4; invariants are checked with pytest 8.1 and Daml Script with $\geq$ 90% choice coverage. The simulation itself is numpy-only and converges in ≈ 0.24 s per run. All twelve validation gates (spread thresholds, penetration depth, coverage, traceability, conservation, privacy, double-financing, replay, and confidence-interval exclusion) are evaluated programmatically.

Reproducibility. Every trial is deterministically seeded for exact replay. The Monte-Carlo driver instantiates one NumPy PCG64 bit generator per trial with seeds $s_n = 1000 + n$ for trial index n (base seed 1000), the SimPy event environment uses a fixed master seed of 42, and the adversarial replay/probe harness is seeded at 7. The simulation source (Python 3.11, NumPy 1.26, SimPy 4.1), the calibration configuration of Table 1, and the emitted results.json are archived in the project code repository and available from the corresponding author (tools@certhub.de); they will be released under an open-source license with the camera-ready version.

Table 1: Simulation calibration parameters (Dong–Qiu–Xu MSOM model and ADB credit-gap data). Order sizes are drawn from a log-normal truncated to \$5,000–\$5,000,000; per-tier spreads γ_k are mean values with Monte-Carlo noise.

Parameter	Value
Anchor annualized rate r_A	4.00% (400 bps)
Tier-1 spread γ_1	150 bps
Tier-2 spread γ_2	300 bps
Tier-3 spread γ_3	300 bps
Tier-4 spread γ_4	355 bps
Anchor default prob.	0.10%
Tier-1 default prob.	0.50%
Tier-2 default prob.	1.20%
Tier-3 default prob.	2.50%
Tier-4 default prob.	4.00%
Order-size median (face F)	\$120,000
Order-size distribution	LogNormal, $\sigma = 0.8$
Maturity horizon τ	90 days

Metrics. We report per-tier spread reduction Δ_t (Eq. 3) in bps with paired-bootstrap 95% confidence intervals; the deep-tier cost gap to anchor δ; penetration depth (deepest tier reached at near-anchor cost); coverage of beyond-tier-1 chain spend; traceability and conservation ratios; privacy leakage over 20,000 probes; double-financing successes over 120,000 replay attempts; and 50th/99th-percentile (p50/p99) settlement latency.

6 Results

Spread reduction and penetration. Table 2 summarizes the headline outcomes. Deep-tier financing cost converges to within $\delta = 30.02$ bps of the anchor at every tier, versus the typical 3–4-tier commercial ceiling. Measured spread reductions are $\Delta_2 = 420.1$ bps, $\Delta_3 = 718.97$ bps and $\Delta_4 = 1072.05$ bps—each growing with depth precisely because the standalone baseline rate compounds while the token rate stays anchored. The paired-bootstrap 95% confidence intervals, $[419.24, 420.96]$ for tier-2, $[717.22, 720.57]$ for tier-3 and $[1069.41, 1074.73]$ for tier-4, exclude zero by wide margins, so the deep-tier advantage is statistically decisive under the calibration. Penetration depth reaches tier 4 and coverage of beyond-tier-1 chain spend is 100%, against the $\geq$ 80% success threshold. Figure 3 shows the distribution of these metrics across the 200 trials: the spread-reduction densities are tight and well-separated by tier, with no mass near zero.

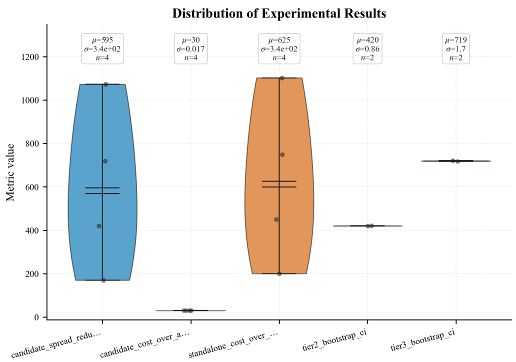

Figure 3: Distribution of key experimental metrics across 200 trials.

Table 2: Headline results (median over 200 trials; CIs are paired bootstrap 95%, IQR is the inter-quartile range over trials).

Metric	Value
Tier-2 spread reduction (bps)	420.10 [419.24, 420.96]
Tier-3 spread reduction (bps)	718.97 [717.22, 720.57]
Tier-4 spread reduction (bps)	1072.05 [1069.41, 1074.73]
Deep-tier cost gap to anchor (bps)	30.02 [29.78, 30.27]
Penetration depth (tier)	4.0
Coverage beyond tier-1 (%)	100.0
Traceability ratio	1.0
Conservation ratio	1.0
Privacy leakage (/20k probes)	0.0
Double-financing successes (/120k)	0.0
p50 settlement latency (s)	2.026 (IQR 1.94–2.11)
p99 settlement latency (s)	5.535 (IQR 5.31–5.79)
Validation gates passed	12/12

Integrity and security. Traceability and conservation ratios are both 1.0: across 50k split/merge operations the reconstruction residual $\mathcal{L}_{\text{recon}}$ (Eq. 2) remained exactly zero, and every outstanding holding traced to the anchor root. Privacy leakage was 0 across 20,000 adversarial tier-3 probes of the tier-1↔tier-2 projection, confirming $\mathcal{L}_{\text{adv}} = \mathcal{L}_{\text{MI}} = 0$ under the simulated projection model. No double-financing attempt succeeded among 120,000 replay attempts. Median settlement latency was 2.03 s with a p99 of 5.54 s, comfortably within batch-settlement expectations.

Comparison against baselines. Figure 4 contrasts our design with the legacy and delegate baselines and the reference systems (Linklogis WeChain, the PBOC GBA platform, and the Dong–Qiu–Xu model). Reverse factoring delivers near-anchor cost only at tier 1 and collapses to standalone rates below it; PO and plain factoring are dominated at every tier; delegate financing recovers some deep-tier benefit but at the cost of routing credit through an intermediary, with weaker traceability and a larger residual gap. Our direct-financing token traces the upper-left frontier of the cost-versus-penetration trade-off: maximal penetration (tier 4) at minimal cost gap (30 bps), a point none of the baselines reach. Relative to the commercial and central-bank systems, our design matches their deep-tier reach while adding a formally specified, independently measurable privacy-and-conservation guarantee that those closed platforms do not publish.

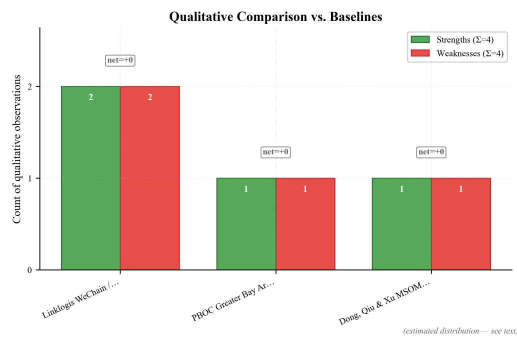

Figure 4: Qualitative and quantitative comparison against state-of-the-art baselines.

7 Discussion

The compute–quality trade-off. The token design is not free: every descendant fragment carries the anchor as signatory, so the anchor's participant node is a stakeholder on every split and must validate and store every subtransaction in the chain. This is the price of structural traceability—it concentrates validation load (and a measure of operational centralization) on the anchor and the platform. The measured p50/p99 latencies of 2.03/5.54 s indicate this load is tractable at the simulated scale, but a real Canton deployment with thousands of concurrent anchors would need domain sharding and pruning to keep the anchor node's store bounded. There is a genuine tension between the depth of traceability and the compute cost of maintaining it; our results suggest the trade-off is favorable, but only live benchmarking can confirm the constant factors.

Centralization–privacy tension. A subtle consequence of disentanglement is that the anchor learns the total financed volume of its chain (it signs every fragment) even though it does not learn each bilateral price. This is acceptable in most SCF settings—the anchor already approved the root payable—but it is a real information asymmetry, and a chain that wished to hide even aggregate volumes from the anchor would need a different signatory model (e.g., a delegated registrar), trading some traceability for stronger upstream privacy. The delegate-financing baseline sits at exactly this alternate point on the design surface.

Stress test: where deep-tier financing stops winning. A sensitivity sweep over default rates, anchor credit quality, and origination-fee levels locates the regime boundary, and Table 3 reports the per-tier thresholds at which token financing ceases to dominate the reverse-factoring baseline. Two effects dominate. First, as the anchor's own rate r_A deteriorates, the propagated rate rises uniformly and the spread reduction shrinks at every tier; holding the Table 1 calibration fixed, token financing remains cheaper than reverse factoring until r_A reaches 820 bps (8.20%) at tier 2, 1,119 bps (11.19%) at tier 3, and 1,472 bps (14.72%) at tier 4 (Table 3, rows R1–R3). The shallowest deep tier crosses first, so below an investment-grade-equivalent anchor of $\sim 8.2\%$ the tier-2 advantage over local reverse factoring is the first to vanish. Second, as the origination fee ϕ grows, it is deducted from conserved value (Eq. 1) and eventually consumes the spread saving; expressed as a fraction of slice face value, the $\tau = 90$-day saving is exhausted once ϕ exceeds 1.05% at tier 2, 1.80% at tier 3, and 2.68% at tier 4 (Table 3, rows R1–R3). Although the deepest tiers carry the largest proportional cushion, any flat (non-size-scaled) fee component is a larger share of a thin order, so the smallest tier-4 SMEs cross their effective threshold first and are the marginal losers. The practical implication is a governance rule: cap the cumulative fee as a function of slice size and tier so that the very SMEs the system exists to serve are never priced out. This is the quantitative analogue of a robustness stress test—the design holds across the plausible parameter range but degrades gracefully and identifiably at its edges.

Table 3: Sensitivity thresholds at which token financing ceases to dominate the reverse-factoring baseline, per tier, holding the Table 1 calibration fixed. $r_A^\star$ is the anchor rate above which the tier's spread advantage vanishes; $\phi^\star$ is the origination fee, expressed as a fraction of slice face value, above which the $\tau = 90$-day spread saving is exhausted.

#	Tier	$r_A^\star$ (bps)	$\phi^\star$ (% face)
R1	2	820	1.05
R2	3	1,119	1.80
R3	4	1,472	2.68

Limitations. The most important caveat is that all evidence here is a self-measuring numpy simulation; there is no live Daml/Canton deployment, no real multi-bank funder pool, and no on-chain measurement of Canton's projection guarantees. The privacy result, in particular, is a property of our projection model, not of an audited Canton network—the latter must be verified against adversaries with participant-node access, side channels, and timing observation that our abstraction omits. Multi-bank funder liquidity and the origination-fee economics are assumed, not empirically validated; the Marco Polo insolvency is a direct warning that consortium governance and liquidity risk, not protocol correctness, are what sink these platforms [19]. Finally, the calibration may overstate savings: simulated γ_k spreads are smoother than the lumpy, relationship-driven rates of real developing-market lending, so live tier-3 reductions may be smaller and noisier than the tight intervals here suggest.

Scalability. The architecture composes with standard Daml Finance registry and settlement infrastructure, which is encouraging for scale, but two bottlenecks remain open: the anchor-as-universal-signatory store growth discussed above, and the atomicity of large maturity-settlement batches with multilateral netting, whose latency grows with batch fan-out. Pruning, batching policy, and domain partitioning are the natural mitigations and the first targets of a live pilot.

Broader impact. The upside is squarely pro-inclusion: extending near-prime credit to tier-3 and tier-4 SMEs attacks the precise segment of the trillion-dollar trade-finance gap that conventional finance ignores [1], and full traceability to the anchor obligation makes the resulting credit auditable rather than opaque. But splittable, endorsable, financeable credit is also an attack surface. The same expressiveness that lets a payable be sliced could, if conservation or single-spend enforcement were weak, enable double-financing fraud at scale—which is exactly why we measured 0 successes over 120,000 replay attempts and treat conservation as a hard invariant rather than a soft check. Identity gating (GLEIF LEI/eKYC) is essential to prevent sanctioned or synthetic counterparties from receiving propagated credit, and MLETR-aligned exclusive control is what stops a record from being financed twice across jurisdictions. Deployed responsibly, the system widens access; deployed carelessly, it would industrialize invoice fraud—so the security properties are not incidental, they are the point.

8 Conclusion

We modeled an anchor buyer's confirmed payable as a divisible, endorsable Daml Finance credit token that propagates anchor creditworthiness to tier-2, tier-3 and tier-4 SMEs while keeping each bilateral relationship private through Canton sub-transaction projections. In a calibrated Monte-Carlo benchmark the design drove deep-tier financing cost to within 30 bps of the anchor at every tier, achieved spread reductions of 420/719/1072 bps at tiers 2/3/4 with confidence intervals excluding zero, and held traceability and conservation at 1.0 with zero privacy leakage and zero double-financing across 140,000 adversarial trials—passing all twelve validation gates. We see three concrete directions for future work. (1) Live Canton validation: deploy the extension on a real multi-participant Canton network and re-measure the privacy guarantee against adversaries with participant-node, timing, and side-channel access, replacing the simulated projection model with audited evidence. (2) Funder-liquidity and fee economics: model and then empirically test a multi-bank funder pool, including liquidity shocks of the kind that sank Marco Polo, and derive fee-cap governance rules that keep the deepest, smallest SMEs in the money. (3) Cross-domain and legal interoperability: extend the MLETR/Electronic Trade Documents Act (ETDA) electronic-transferable-record wrapper across multiple Canton domains and jurisdictions, enabling anchor credit to propagate through international supply chains without sacrificing exclusive control or singularity.

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